ORGANOMETALLIC CHEMISTRY

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1. **Introduction**

2. **16 and 18 electron rule (learning to count)**

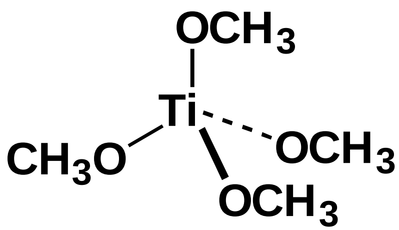
3. **Bonding of selected Organometallic ligands**

4. **Characterization technique: IR**

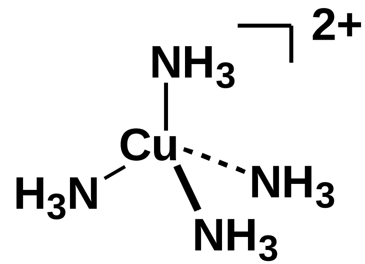
5. **Applications (Catalysis).**

Which of the following compounds is an organometallic compound?

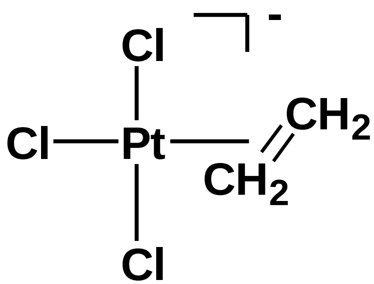
a)



b)



c)



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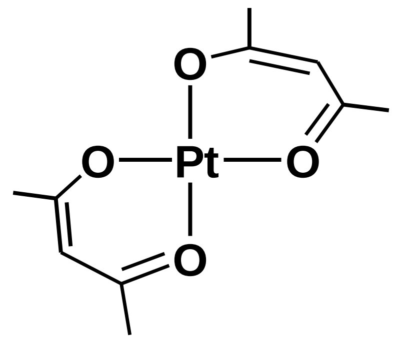
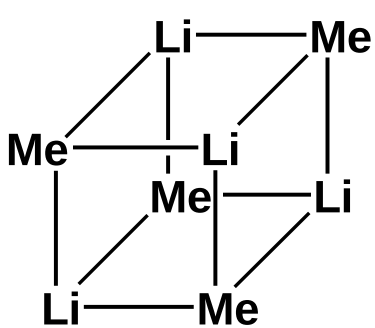
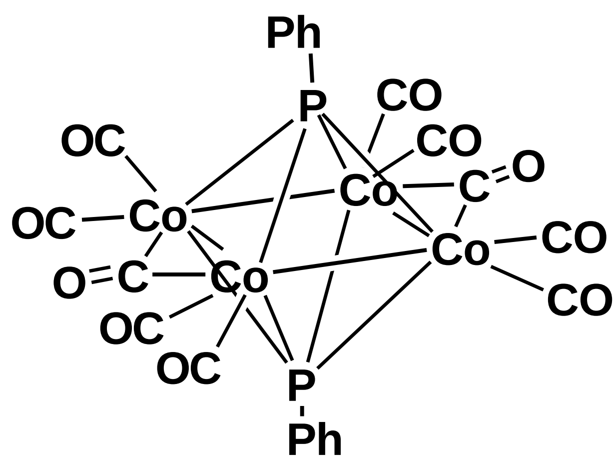


e)



f)



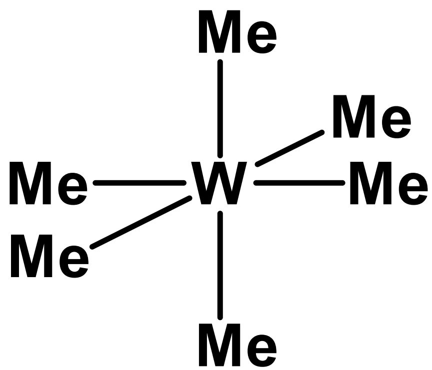
In general, metals in organometallic compounds include:

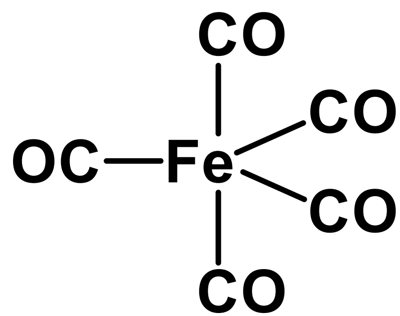
- ***main group metals***
- ***transition metals***
- ***f-block metals***

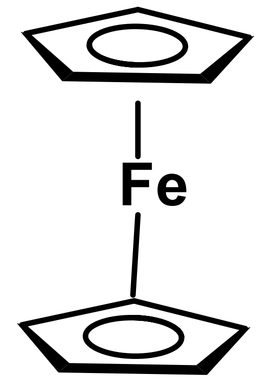
In this course, transition metals are our main concern.

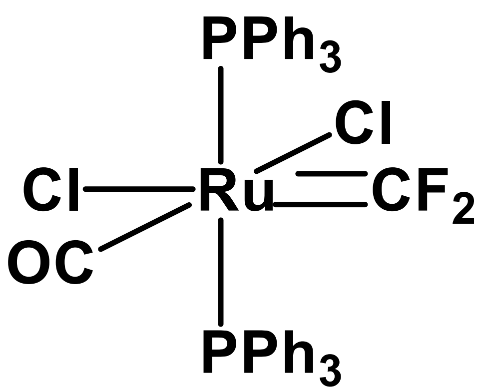
Chemistry: structures, bonding and properties of molecules.

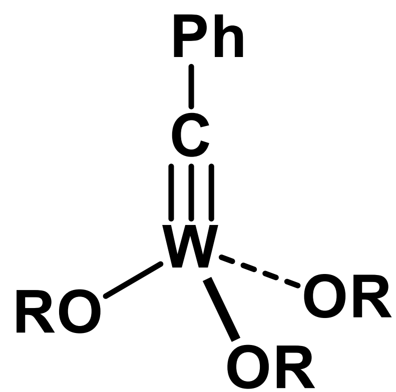
Transition metal compounds









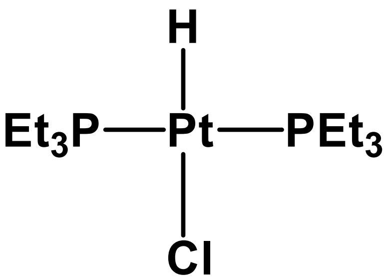


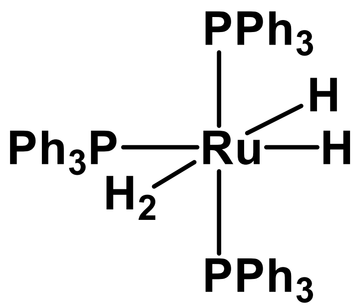
MeLi

PhMgBr

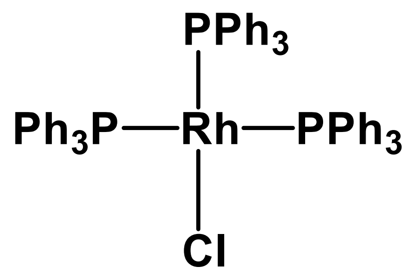
Some compounds do not contain metal-carbon bond, but they are usually included in the field of organometallic chemistry. They include:

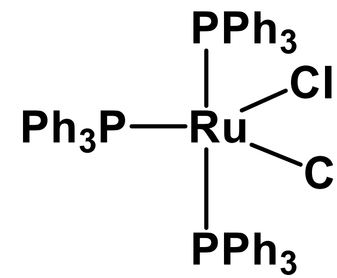
- **Metal hydride complexes, *e.g.***



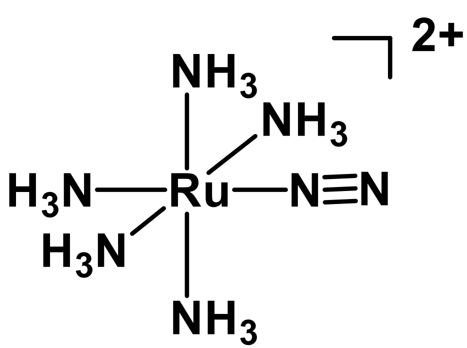


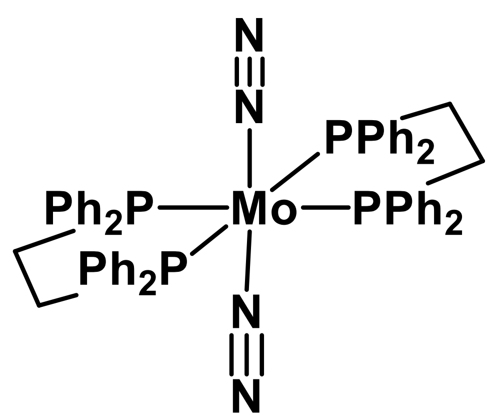
- **Phosphine complexes, *e.g.***





- **N_2-complexes, *e.g.***



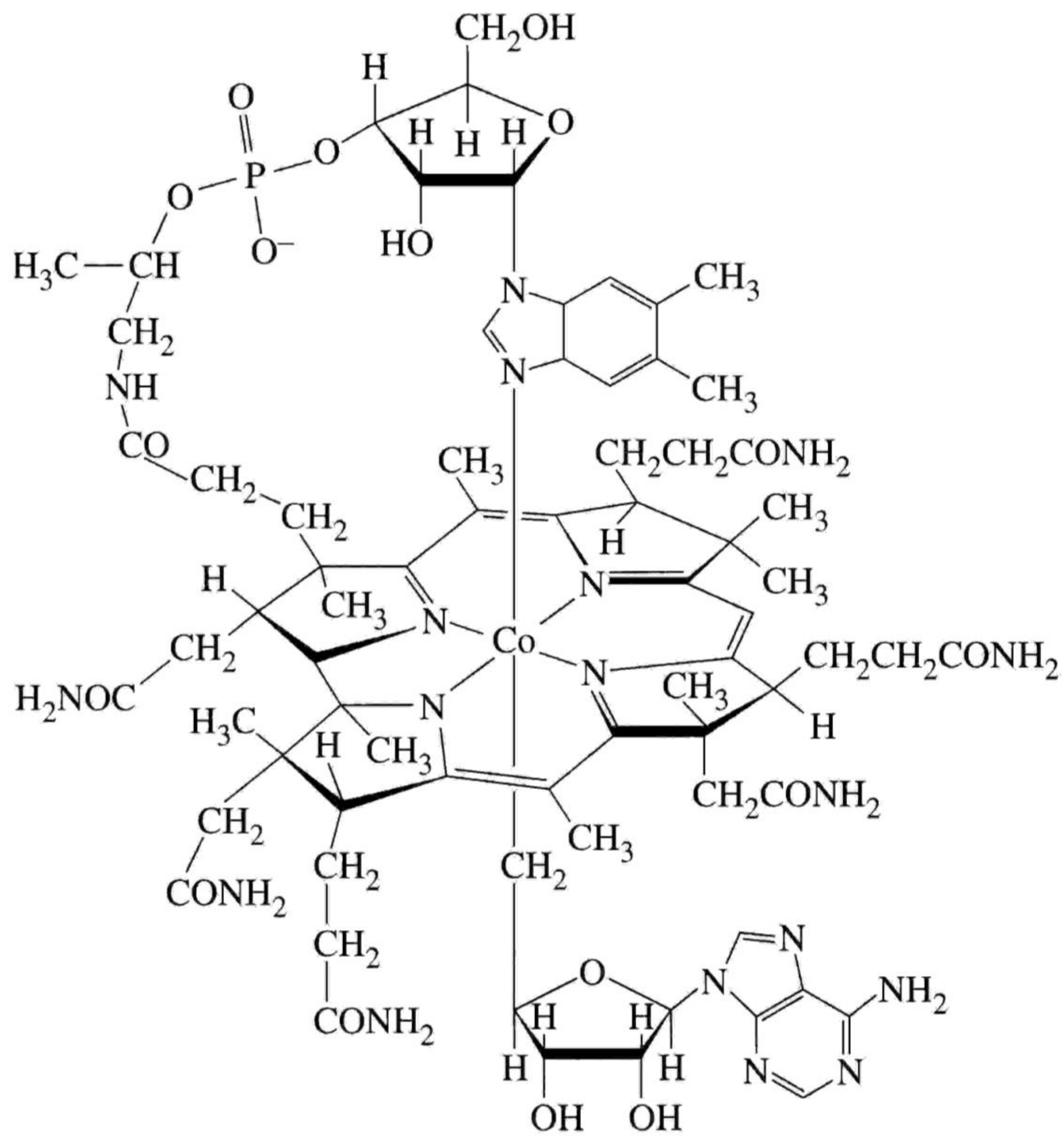


A brief history of organometallic chemistry

1) **Organometallic Chemistry has really been around for millions of years**

   Naturally occurring
   Cobalamins contain
   Co—C bonds

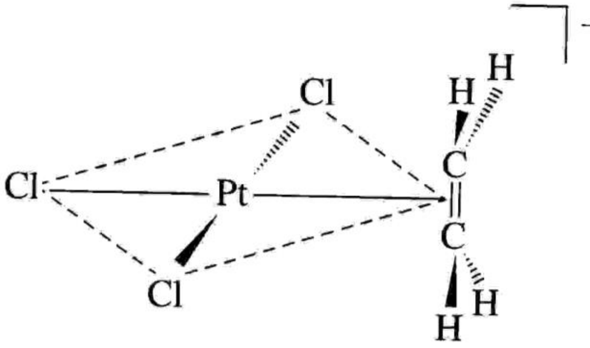
   Vitamin B12



2) Zeise's Salt synthesized in 1827 = $K[Pt(C_2H_4)Cl_3] \cdot H_2O$

Confirmed to have $H_2C{=}CH_2$ as a ligand in 1868

Structure not fully known until 1975



3) $Ni(CO)_4$ synthesized in 1890

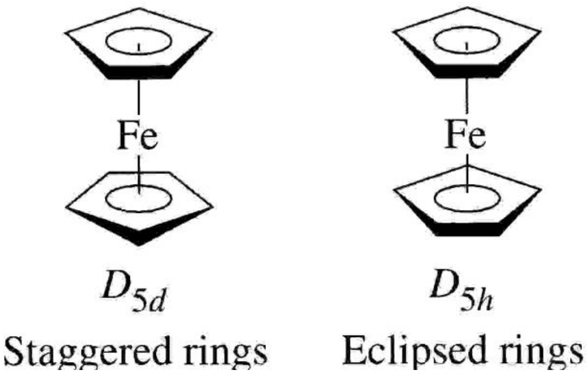
4) Grignard Reagents (XMgR) synthesized about 1900

Utility to Organic Synthesis recognized early on

5) Ferrocene synthesized in 1951

Modern Organometallic Chemistry begins with this discovery (Paulson and Miller)

1952 Fischer and Wilkinson



Nobel -Prize Winners related to the area:

Victor Grignard and Paul Sabatier (1912)
Grignard reagent

K. Ziegler, G. Natta (1963)

Zieglar-Natta catalyst

E. O. Fisher, G. Wilkinson (1973)

Sandwich compounds

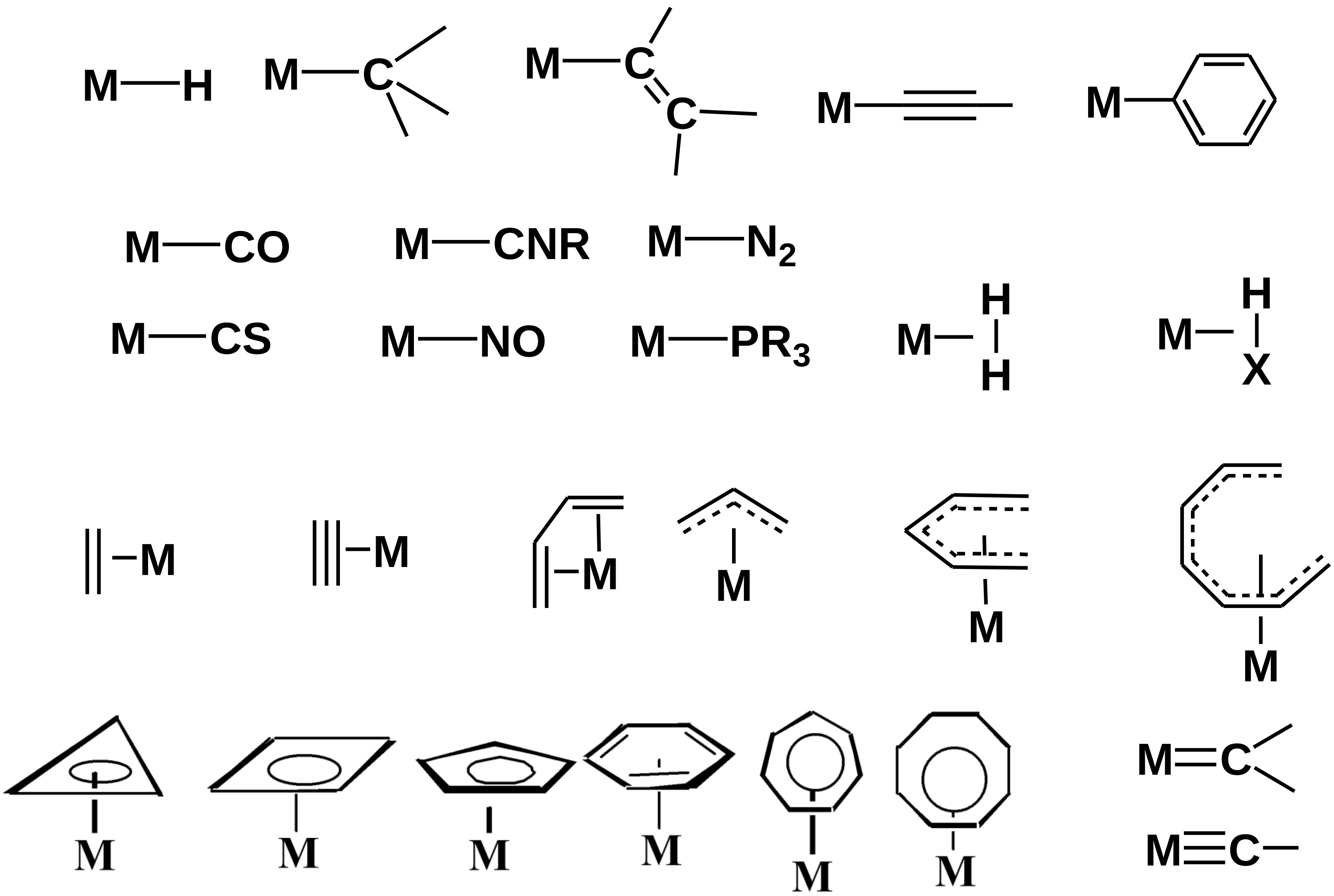
K. B. Sharpless, R. Noyori (2001)

Hydrogenation and oxidation

Yves Chauvin, Robert H. Grubbs, Richard R. Schrock (2005)

Metal-catalyzed alkene metathesis

Common organometallic ligands



Why organometallic chemistry ?

a). From practical point of view:

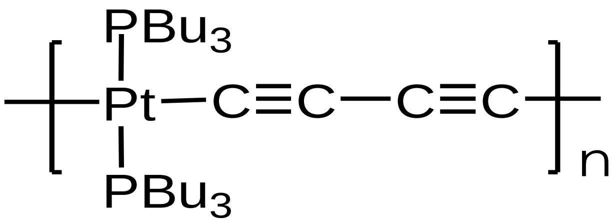
*** OMC are useful for chemical synthesis, especially catalytic processes, *e.g.***

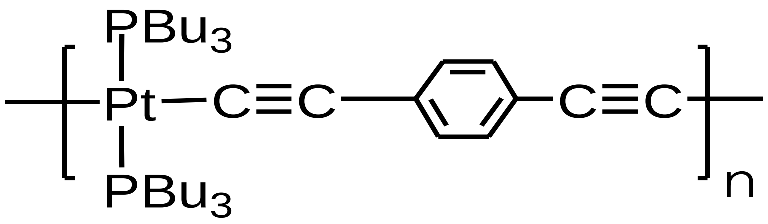
In production of fine chemicals

In production of chemicals in large-scale reactions could not be achieved traditionally

* **Organometallic chemistry is related to material sciences.**

e.g.* *Organometallic Polymers





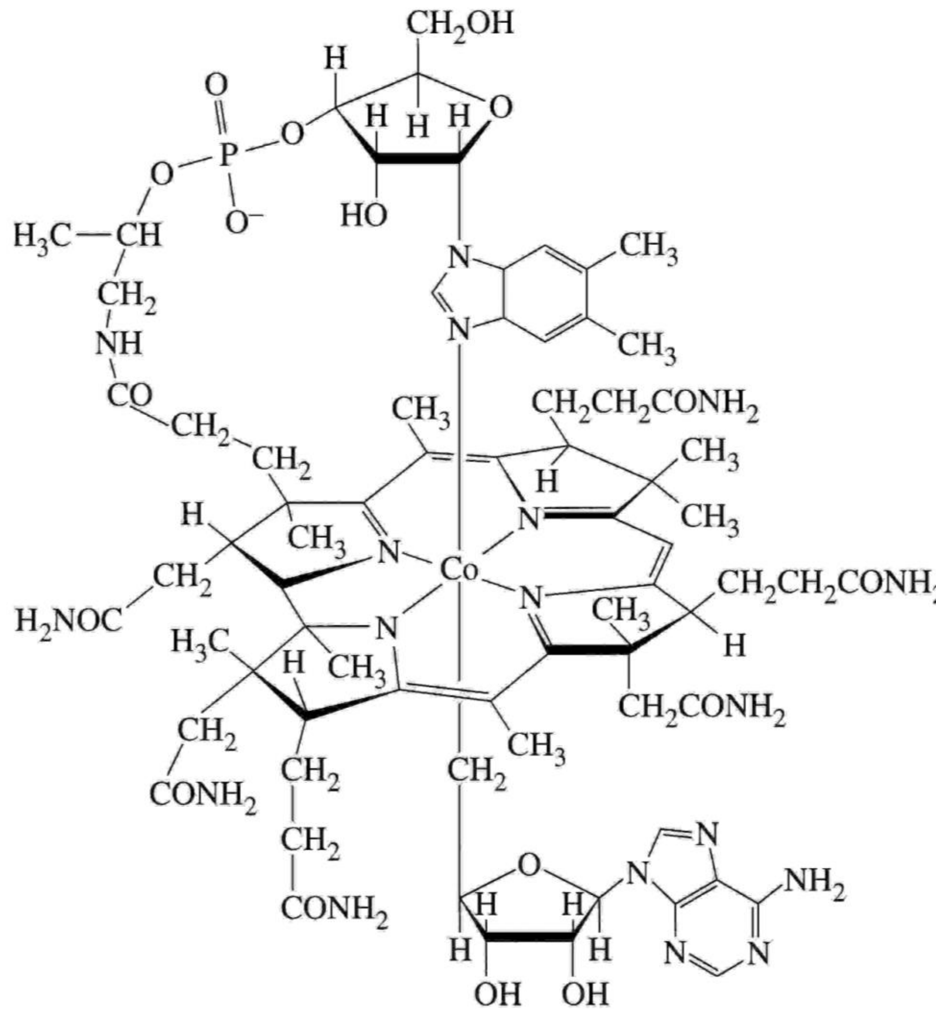
Small organometallic compounds:
Precursors to films for coating (MOCVD)

$CH_3C{\equiv}C$-Au-$C{\equiv}NMe$ -----> Au film

Luminescent materials

*** Biological Science. Organometallic chemistry may help us to understand some enzyme-catalyzed reactions.**

e.g. ***B_{12} catalyzed reactions.***



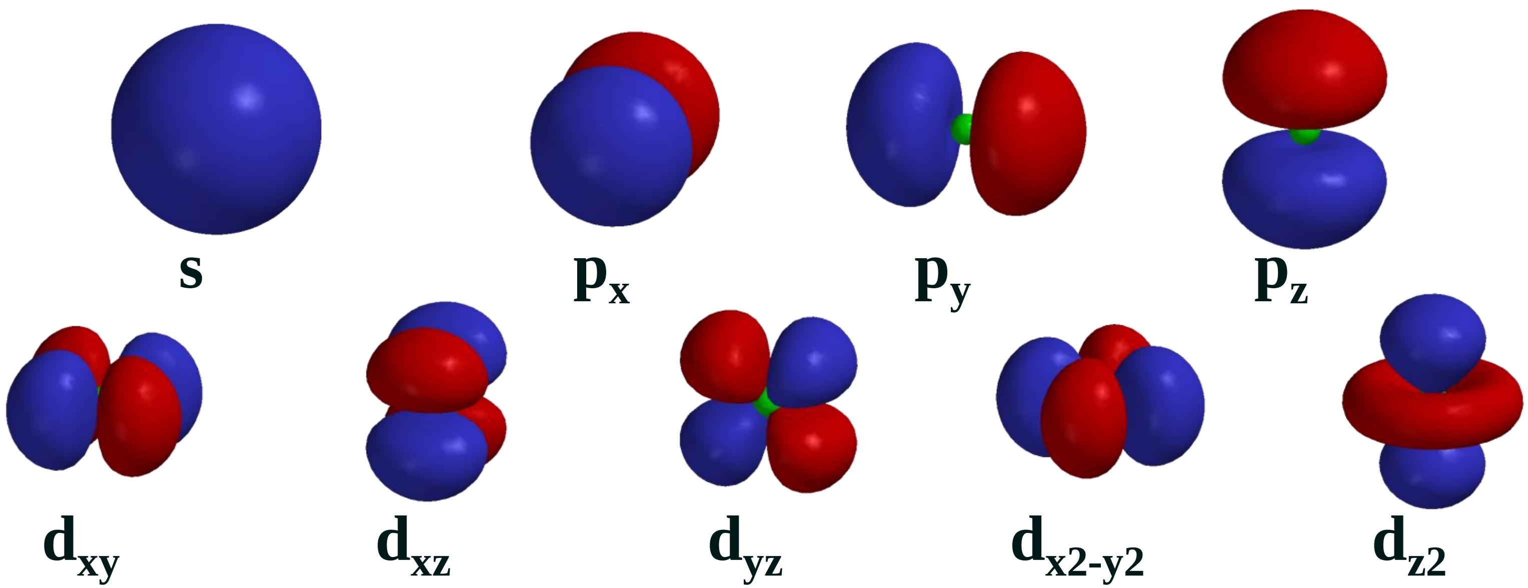
18 Electron Rule (Sidgwick, 1927)

- OM chemistry gives rise to many "stable" complexes - how can we tell by a simple method
- Every element has a certain number of valence orbitals:

  1 { 1s } for H

  4 { *n*s, 3´ *n*p } for main group elements

  9 { *n*s, 3´ *n*p, 5´ (*n*-1)d } for transition metals

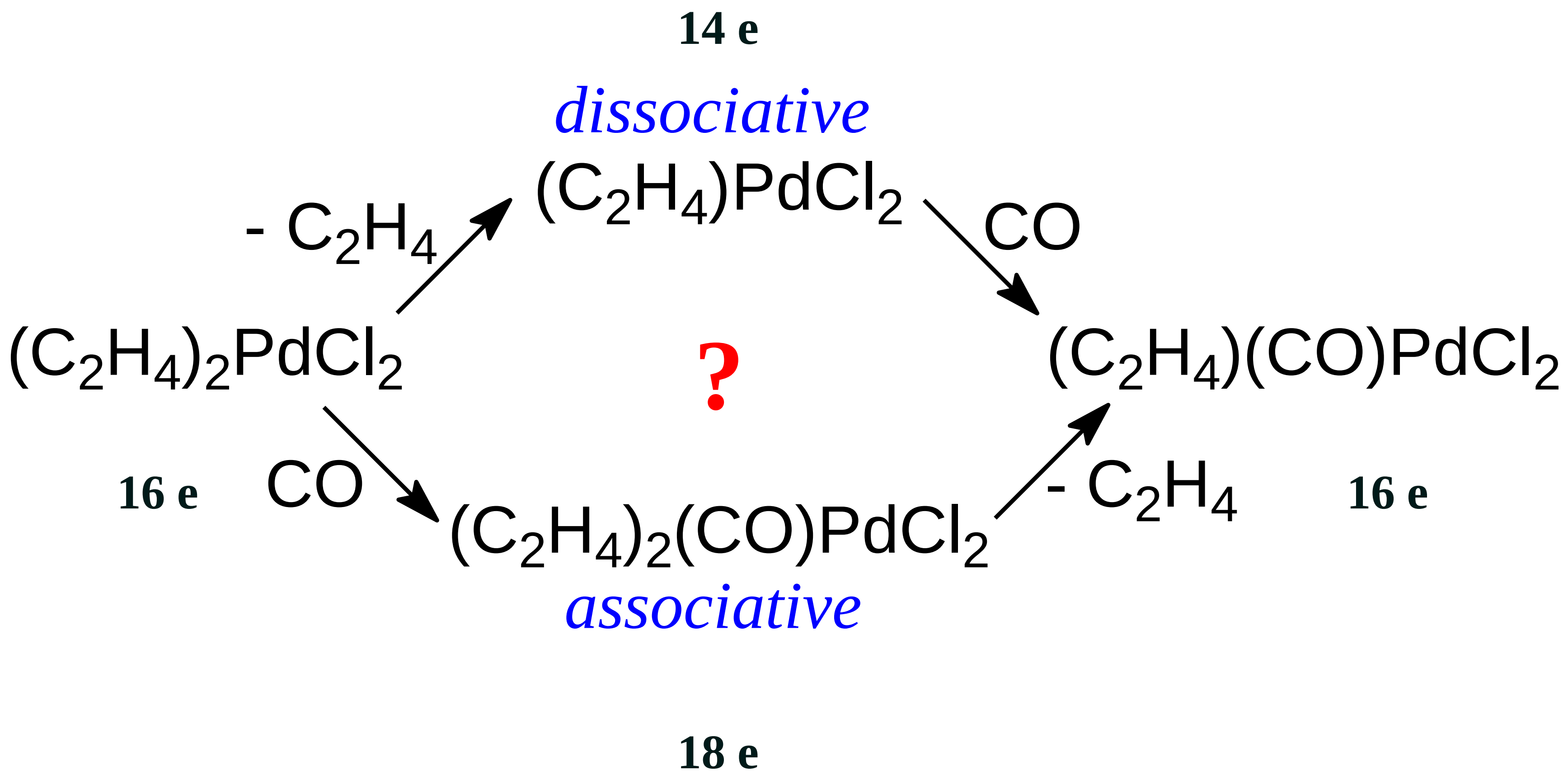


- **Therefore, every element wants to be surrounded by 2/8/18 electrons**
  - **For main-group metals (8-*e*), this leads to the standard Lewis structure rules**
  - **For transition metals, we get the 18-electron rule**

The strength of the preference for electron-precise structures depends on the position of the element in the periodic table

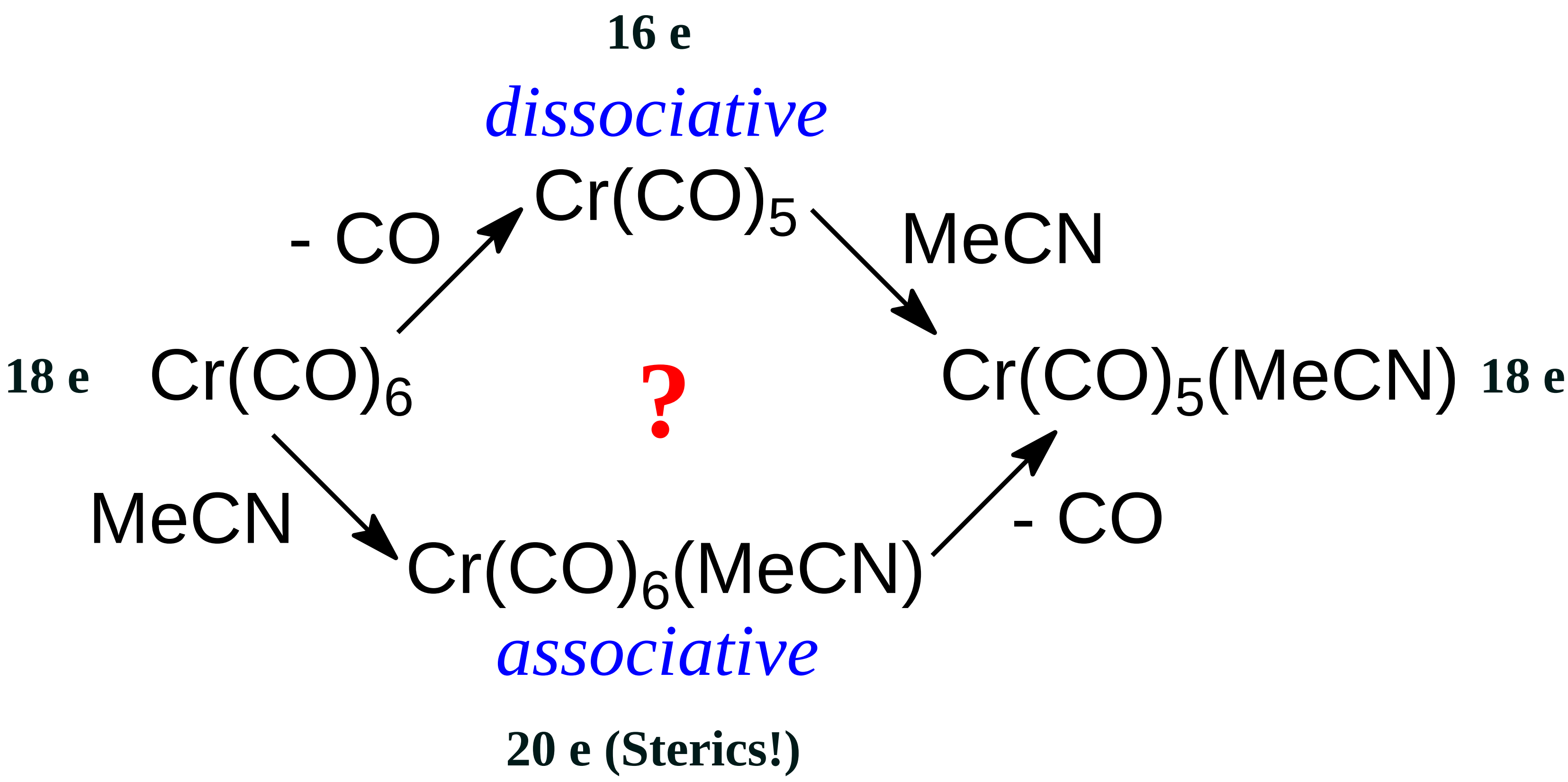
- **For early transition metals, 18-*e* is often unattainable for steric reasons - the required number of ligands would not fit**
- **For later transition metals, 16-*e* is often quite stable (square-planar d^8 complexes)**

Predicting reactivity

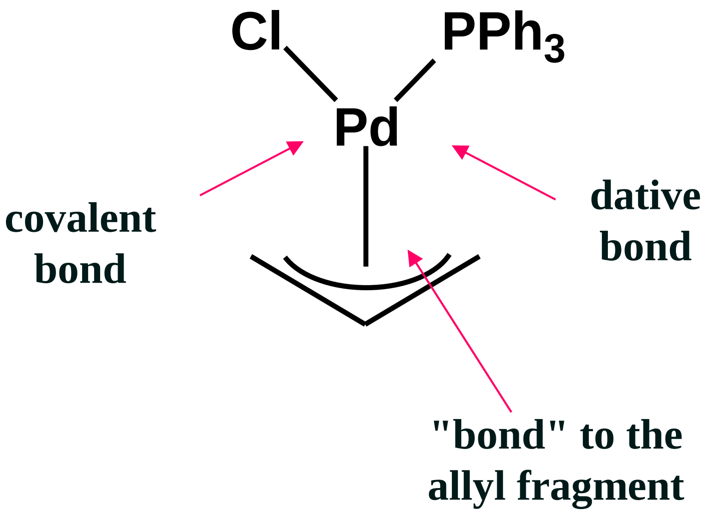


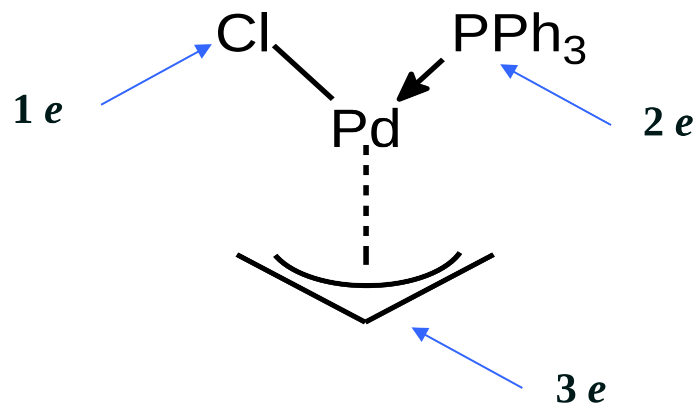
Most likely *associative*

Predicting reactivity



Most likely *dissociative*





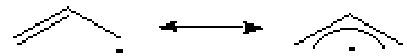
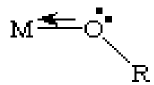
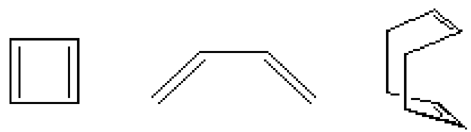
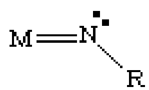
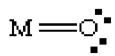
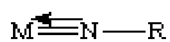
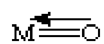
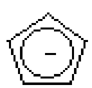
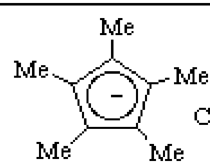
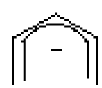
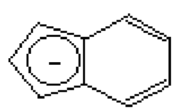
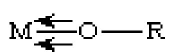
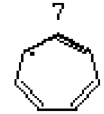
Pd =	10
Cl =	1
PR_3 =	2
allyl =	3
	=
e-count	16

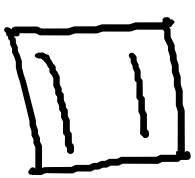
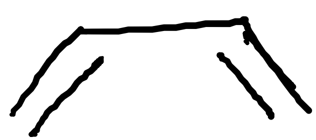
"*Covalent" count*: (ionic method also useful)

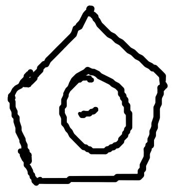
1. **Number of valence electrons of central atom.**
   - **from periodic table**
2. **Correct for charge, if any**
3. **Count 1 *e* for every *covalent* bond to another atom.**
4. **Count 2 *e* for every *dative* bond *from* another atom.**

5. **Delocalized carbon fragments: usually 1 *e* per C (hapticity)**
6. **Three- and four-center bonds need special treatment**

7. **Add everything**

	Neutral/Covalent	Oxid State/Ionic
R (H, Me, Et, Pr, H) -CN, OH, Cl, OR (sp^3), acyl, NO (bent)	1	2
PR_3, amines, ethers, NH_3, R_2S, CO, RCN, RNC, olefins, acetylenes (sometimes), ketones	2	2
 (allyl)  (sp^2 alkoxide)	3	4
NO (linear)	3	2 (NO^+)
 (1,5-COD) Acetylenes (sometimes)	4	4
  (bent imido) (oxo)	2	4
  (linear imido) (oxo)	4	6
 Cp  Cp*   (Indenyl)  (linear alkoxide)	5	6
Arenes (benzene etc.)	6	6
C_7H_7	7 	6 (as a cation) 8 (as an anion)
cyclooctatetraene (COT)	8 (max)	10 (as 2-)

Ligand	Electrons
R (H, Me, Et, Pr) CN, OH, Cl, OR, NO (bent)	1
PR_3, Amines, Ethers, NH_3 R_2S, CO, RCN, olefins,	2
NO (linear)	3
 	4



Cp

5

μ–X   M–X–M   3

X = Halogen

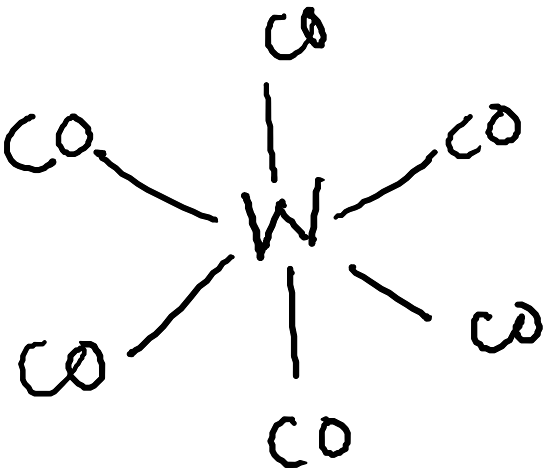
μ-alcoxide

M–OR–M   3

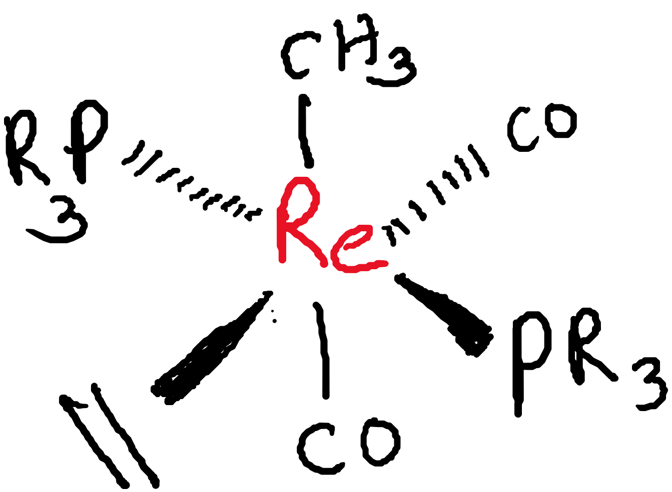
Hapto (h) Number (hapticity)

For some molecules the molecular formula provides insufficient information with which to classify the metal carbon interactions

The hapto number (h) gives the number of carbon (conjugated) atoms bound to the metal



$$\begin{aligned} W &= 6e \\ 6\,CO &= 12e \\ \hline & \ 18e \end{aligned}$$



$$\begin{aligned} Re(+1) &= d^6 \\ 2PR_3 &= 4e \\ 2\,CO &= 4e \\ CH_3^- &= 2e \\ CH_2{=}CH_2 &= 2e \\ \hline & \ 18e \end{aligned}$$

$Fe(CO)_4 PPh_3$

$Fe = d^8$

$4CO = 8e$

$PPh_3 = 2e$

$\overline{\quad 18e \quad}$

$Ni(PPh_3)_4$

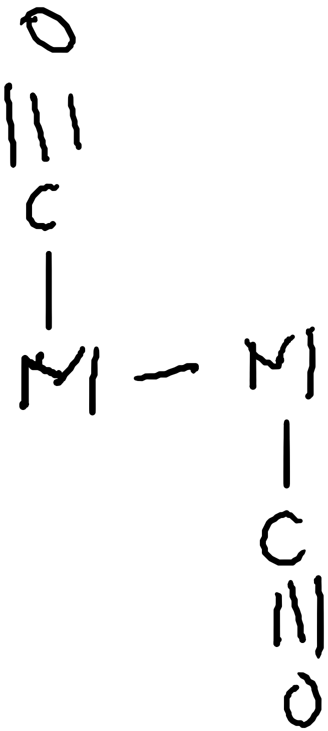
$Ni = d^{10}$

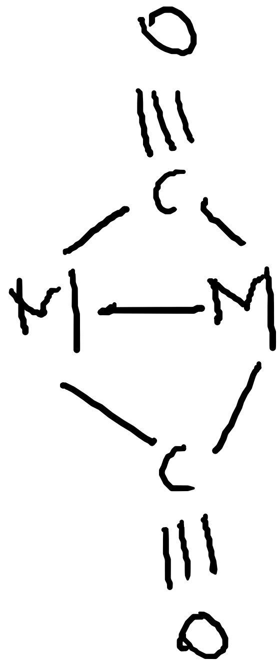
$4PPh_3 = 8e$

$\overline{\quad 18e \quad}$

$Mn(CO)_5$ $Co(CO)_4$

$[Mn(CO)_5]^-$ $[Co(CO)_4]^-$





$Co_2(CO)_8$

$2\,Co = 18\,e^-$

$8\,CO = 16\,e^-$

$Co - Co = 2\,e^-$

$\overline{\quad 36\,e^- \quad}$

$18\,e^-/Co$

Exceptions to the 18 Electron Rule

$ZrCl_2(C_5H_5)_2$ Zr(4) + [2 x Cl(1)] + [2 x C_5H_5(5)] =16

$TaCl_2Me_3$ Ta(5) + [2+ x Cl(1)] + [3 x M(1)] =10

WMe_6 W(6) + [6 x Me(1)] =12

$Pt(PPh_3)_3$ Pt(10) + [3 x PPh_3(2)] =16

$IrCl(CO)(PPh_3)2$ Ir(9) + Cl(1) + CO(2) + [2 x PPh_3(2)] =16

What features do these complexes possess?

- **Early transition metals (Zr, Ta, W)**
- **Several bulky ligands (PPh_3)**
- **Square planar *d*8 e.g. Pt(II), Ir(I)**
- **σ-donor ligands (Me)**

Bonding of selected Organometallic ligands

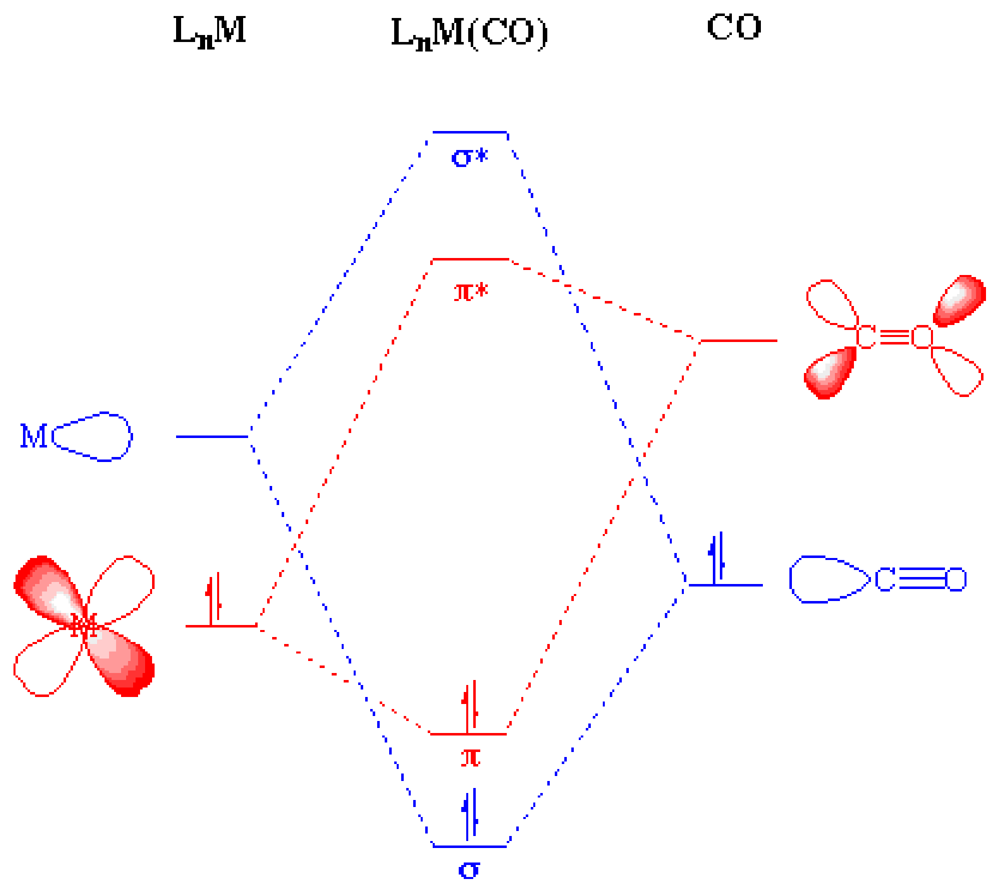
Carbonyl Ligand: CO

Carbonyl Complexes

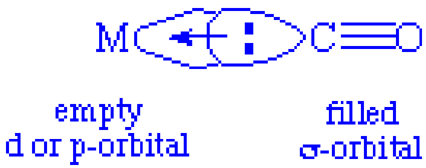
Bonding of CO

Electron donation of the lone pair on carbon. This electron donation makes the metal more electron rich - compensate for this increased electron density, a filled metal *d*-orbital may interact with the empty *p orbital on the carbonyl ligand**

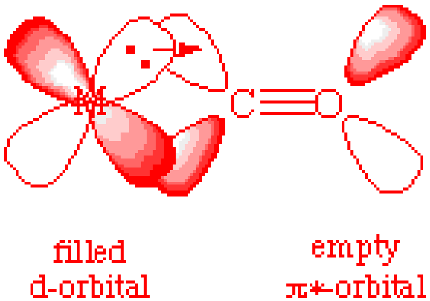
p-backbonding or p-backdonation or synergistic bonding



σ bond:



π backbond:



What stabilizes CO complexes is M→C π–bonding

The lower the formal charge on the metal ion the more willing it is to donate electrons to the π–orbitals of the CO

IR spectra and metal-carbon bonds

IR spectra and metal-carbon bonds

The υ_{CO} stretching frequency of the coordinated CO is very informative

Recall that the stronger a bond gets, the higher its stretching frequency

M=C=O (C=O is a double bond) canonical structure

Lower the υ_{CO} stretching frequency as compared to the M-C≡O structure (triple bond)

Note: υ_{CO} for free CO is 2041 cm^{-1})

IR spectra and metal-carbon bonds

υ_{CO} for free CO is 2041 cm^{-1})

	$[Ti(CO)_6]^{2-}$	$[V(CO)_6]^{-}$	$[Cr(CO)_6]$	$[Mn(CO)_6]^{+}$	$[Fe(CO)_6]^{2+}$
υ_{CO}	1748	1858	1984	2094	2204 cm^{-1}

$\longleftarrow$ increasing M=C double bonding

$\longrightarrow$ decreasing M=C double bonding

Bridging versus terminal carbonyls

Bridging CO groups can be regarded as having a double bond C=O group, as compared to a terminal C≡O, which is more like a triple bond:

~ triple bond

M-C≡O

terminal carbonyl
(~ 1850-2125 cm^{-1})

~ double bond

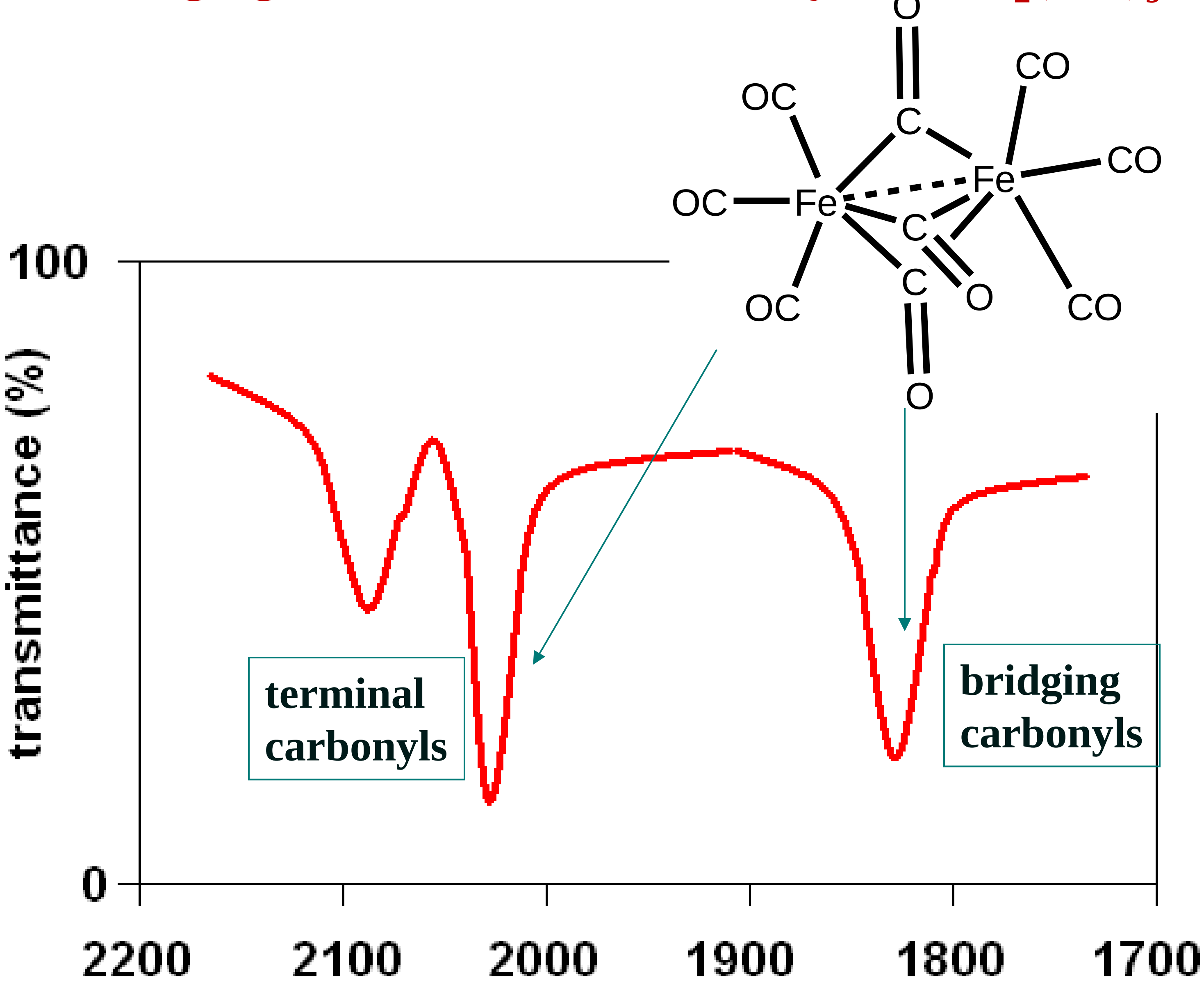
M
C=O
M

bridging carbonyl
(~1700-1860 cm^{-1})

the C=O group in a bridging carbonyl is more like the C=O in a ketone, which typically has $\upsilon_{C=O}$ = 1750 cm^{-1}

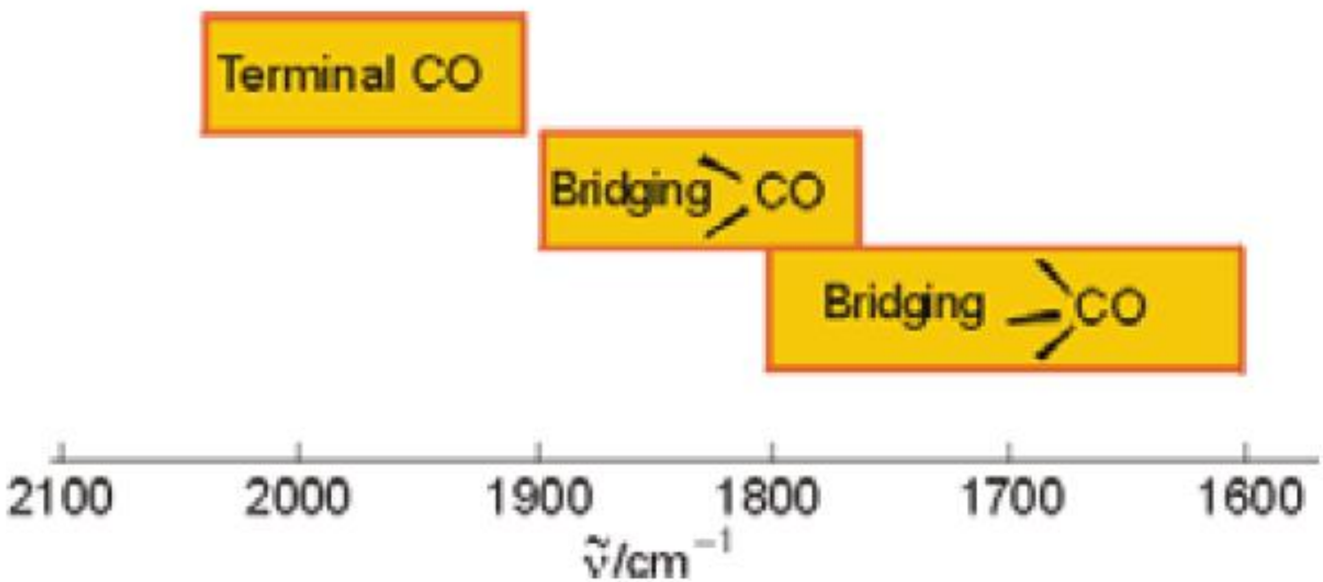
Bridging CO between 1700 and 2200 cm^{-1}

Bridging versus terminal carbonyls in $[Fe_2(CO)_9]$



Summary

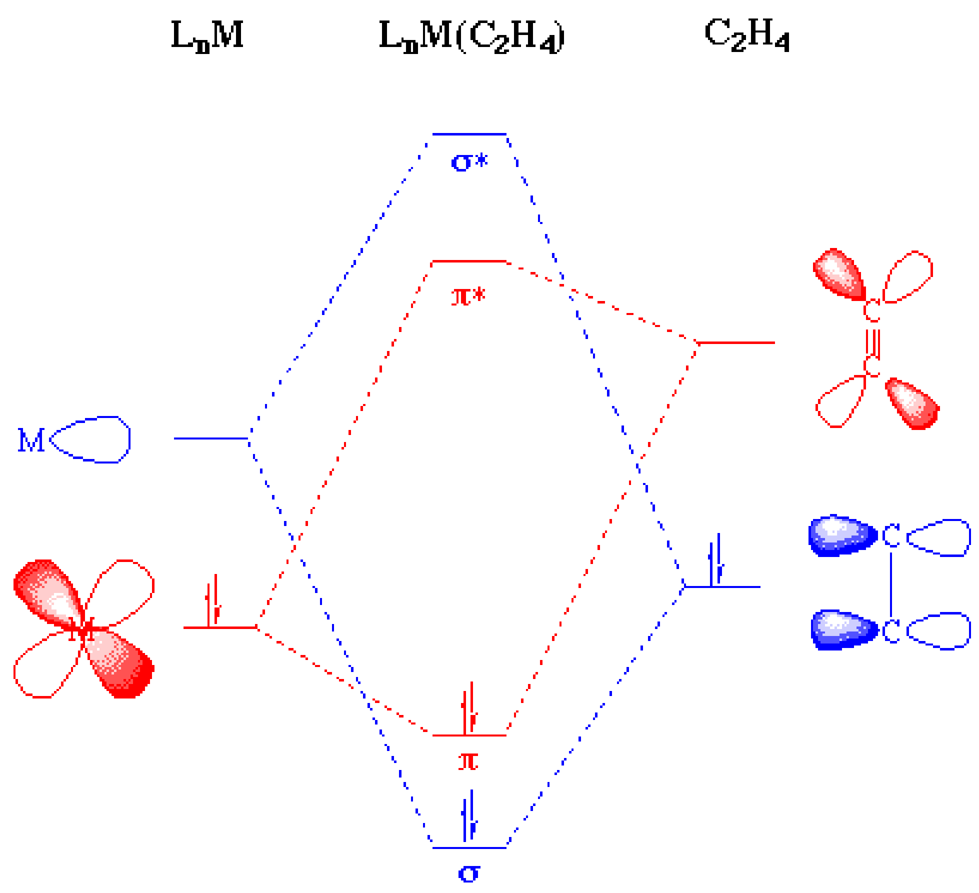
1. As the CO bridges more metal centers its stretching frequency drops – same for all pi ligands – More back donation



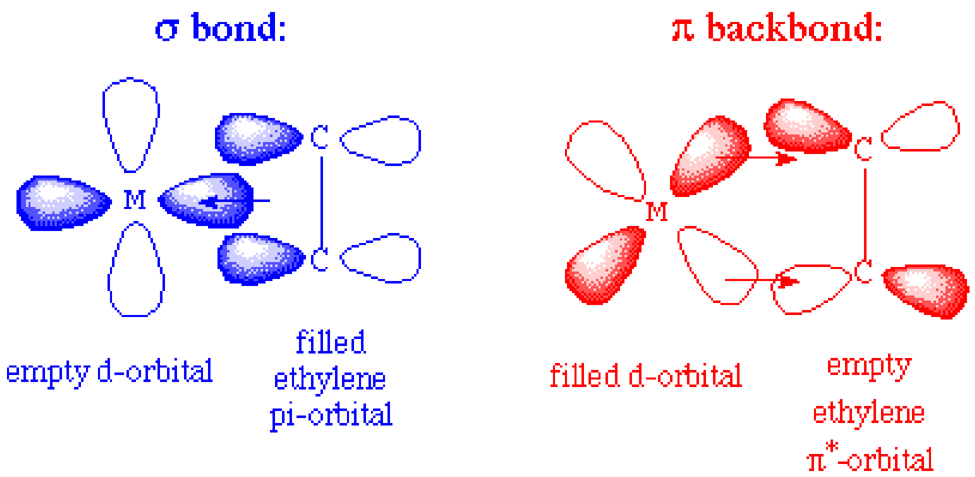
2. As the metal center becomes increasingly electron rich the stretching frequency drops

The influence of coordination and charge on CO stretching frequency	
Compound	Frequency (cm^{-1})
$CO_{(g)}$	2143
$[Mn(CO)_6]^+$	2090
$[Cr(CO)_6]$	2000
$[V(CO)_6]^-$	1860
$[Ti(CO)_6]^{2-}$	1750

Alkene ligands



L_nM
$L_nM(C_2H_4)$
C_2H_4
σ*
π*
M
π
σ
C
C
C
C

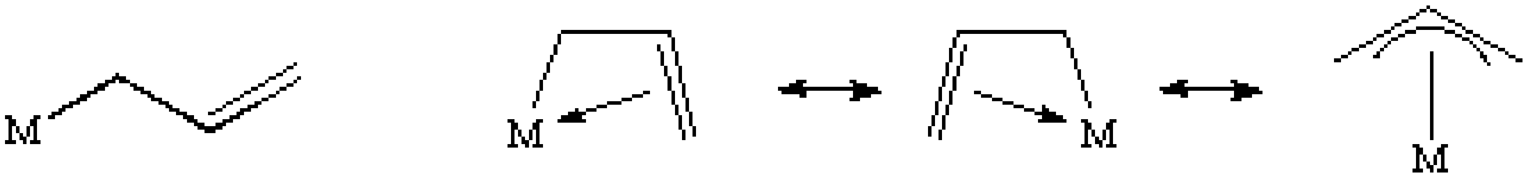


The greater the electron density back-donated into the p* orbital on the alkene, the greater the reduction in the C=C bond order

Stability of alkene complexes also depends on steric factors as well

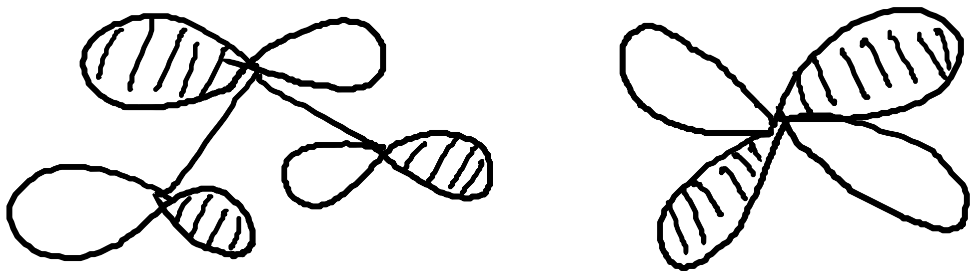
Ally ligands:

Allyl ligands are ambidentate ligands that can bind in both a monohapto and trihapto form The trihapto form can be expressed as a number of difference resonance forms as shown here for an unsubstituted allyl ligand: Important applications



η^1-form η^3-form

A.

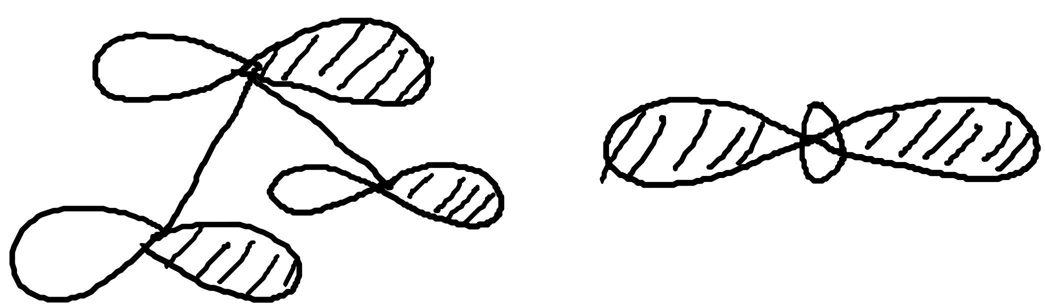


d_{xz}

N.

d_{yz}

B.



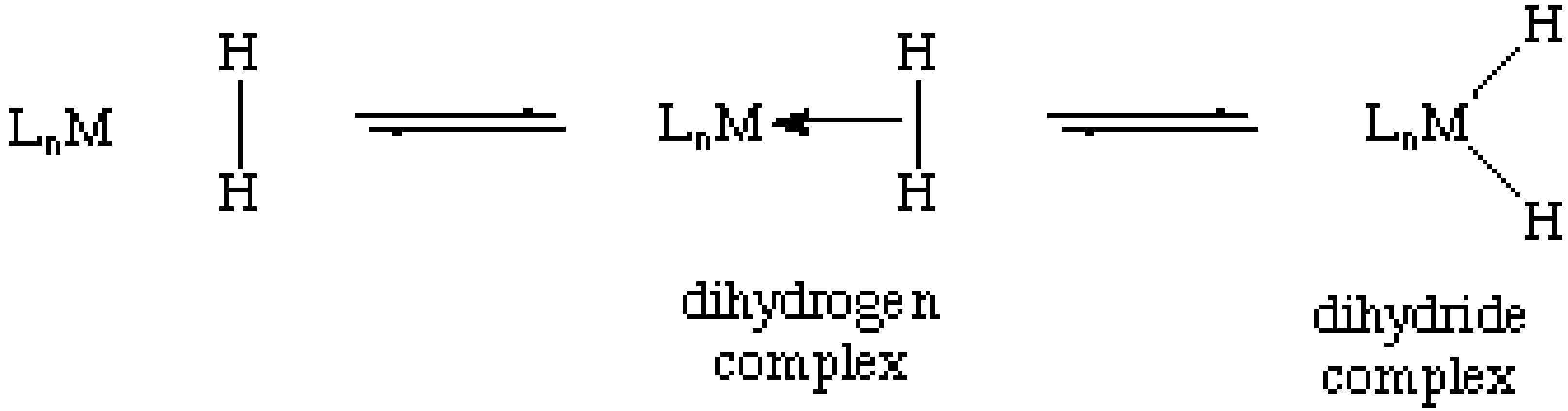
d_{z^2}

Dihydrogen Ligand: H_2

Dihydrogen Ligands:

Metal is more electropositive than hydrogen

Hydrogen acts as a two electron sigma donor to the metal center.

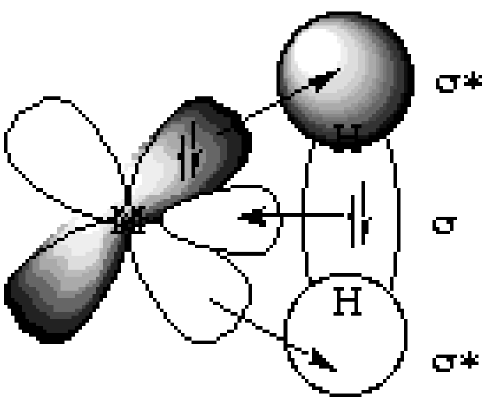


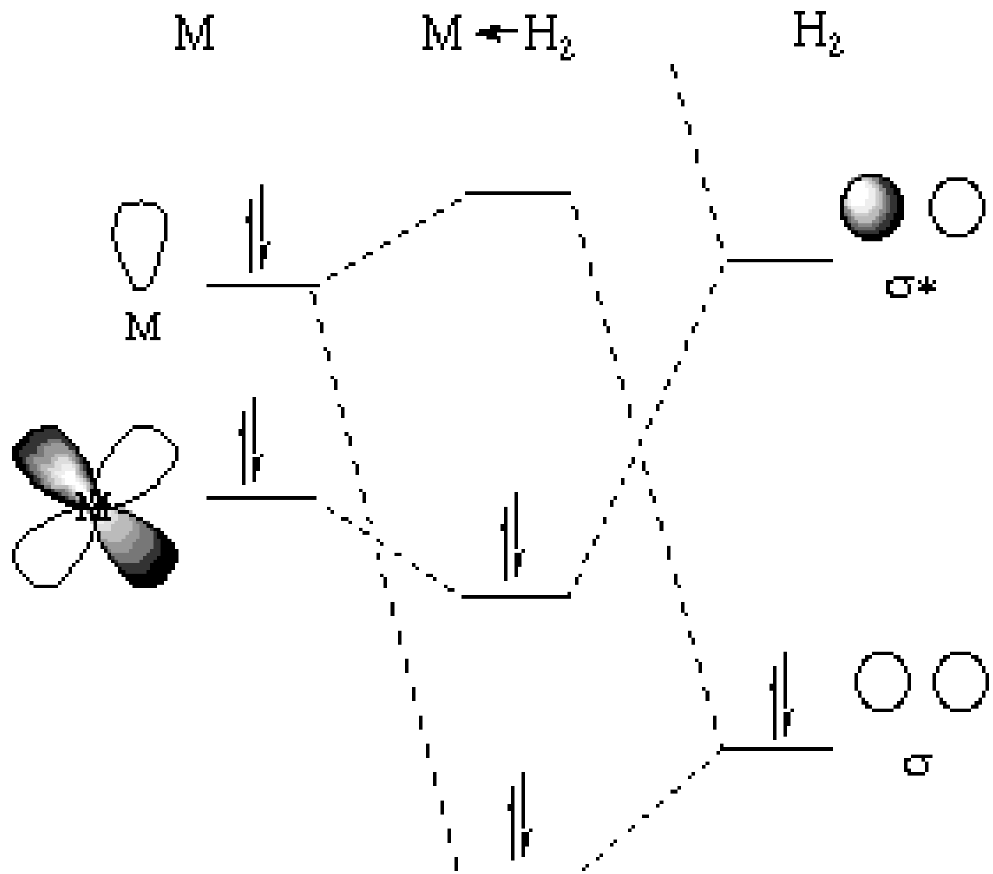
How does this affect the oxidation state of the metal?

Dihydrogen complexes Bonding

H_2 - neutral two electron sigma donor

One could also describe a back-donation of electrons from a filled metal orbital to the sigma-* orbital on the dihydrogen





Reaction chemistry of Organometallic complexes

Reaction chemistry of complexes

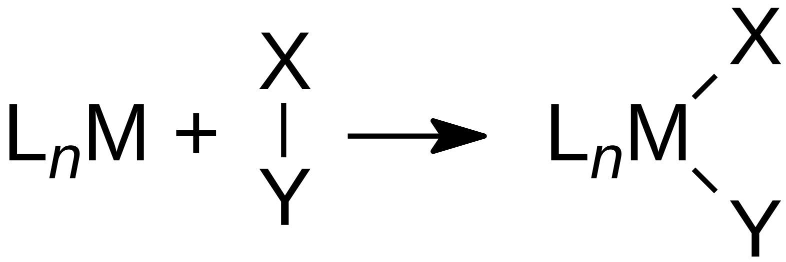
Three general forms:

1. **Reactions involving the gain and loss of ligands**
    - **a. Oxidative Addition**
    - **b. Reductive Elimination**

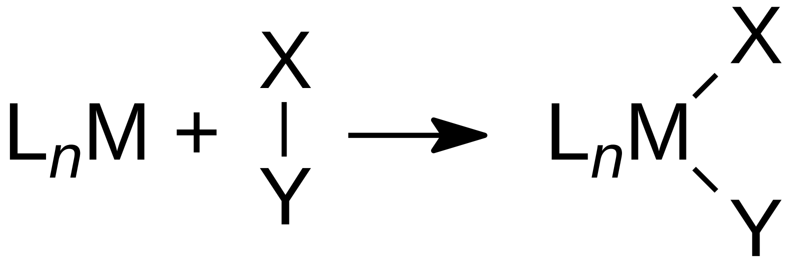
2. **Reactions involving modifications of the ligand**
    - **a. Carbonyl insertion (alkyl migration)**

3. **Catalytic processes by the complexes**
    - **Wilkinson**
    - **Monsanto**
    - **Ziegler Natta catalyst**

a) Oxidative Addition



a) Oxidative Addition



- **Oxidation state changes by + 2**

- **The new M-X and M-Y bonds are formed using:**

  - the electron pair of the X-Y bond

  - one metal-centered lone pair

b) Reductive elimination

This is the reverse of oxidative addition - Expect *cis* elimination

Rate depends strongly on types of groups to be eliminated.

Usually easy for:

- **H + alkyl / aryl / acyl**

Modifications of the ligand

a) Insertion reactions *Migratory* insertion!

The ligands involved must be *cis*

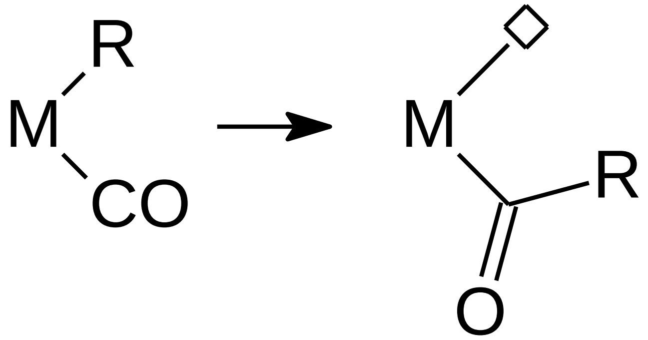
If at a metal center you have a sigma-bound group (hydride, alkyl, aryl) and a ligand containing a pi-system (olefin, alkyne, CO) the sigma-bound group can *migrate* to the pi-system

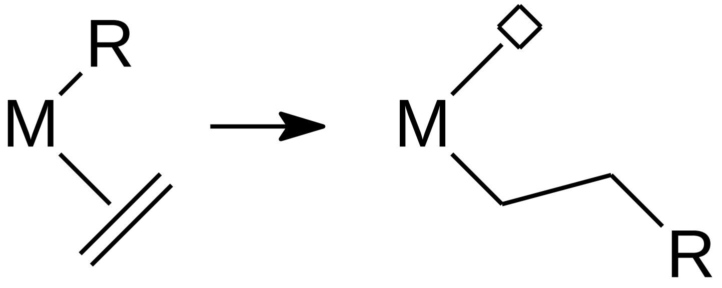
Modifications of the ligand

a) Insertion reactions   *Migratory* insertion!

1. CO : 1,1-insertion

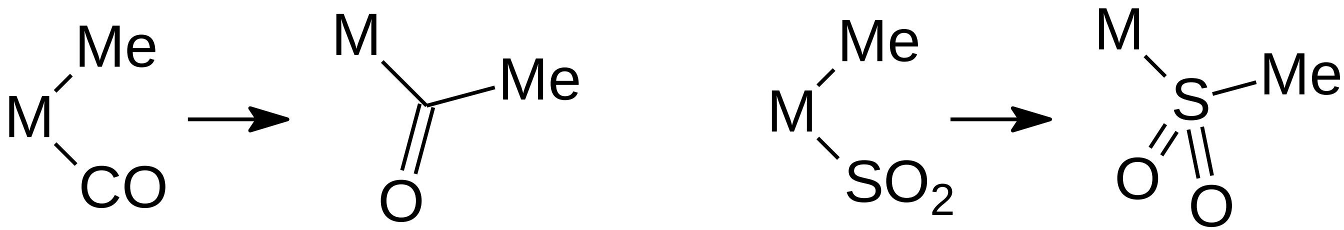
2. Olefins: 1,2-insertion,





In a 1,1-insertion, metal and X group "move" to the same atom of the inserting substrate.

The metal-bound substrate atom increases its valence



CO, isonitriles (RNC) and SO_2 often undergo 1,1-insertion

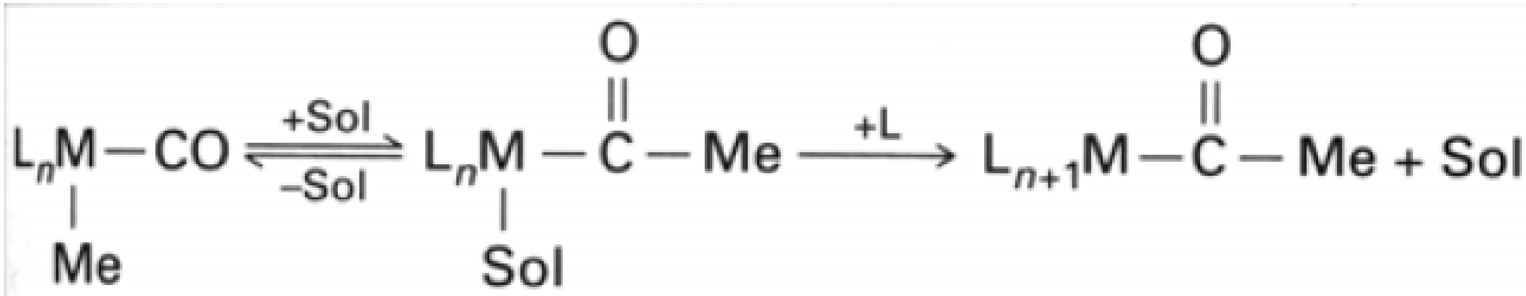
1,2 insertion (olefins)

Insertion of an olefin in a metal-alkyl bond produces a new alkyl

Thus, the reaction leads to *oligomers* or *polymers* of the olefin

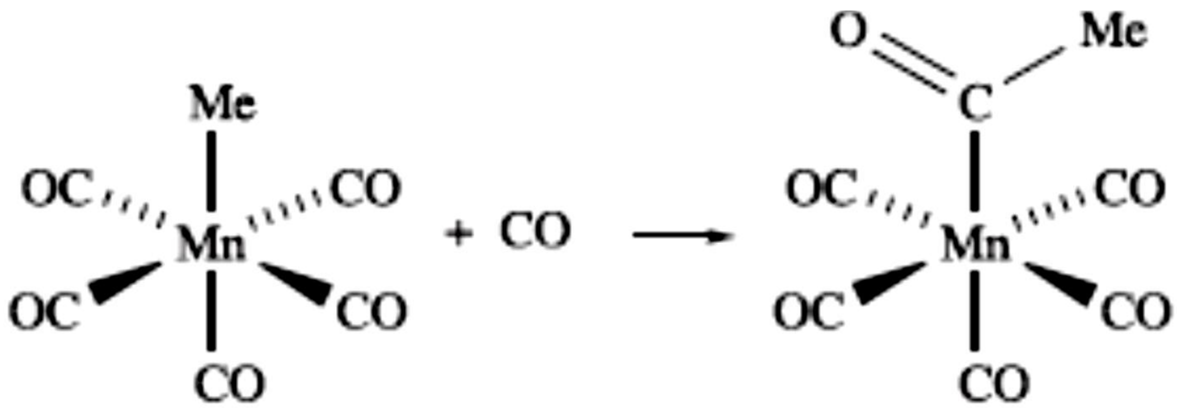
- **polyethene (polythene)**

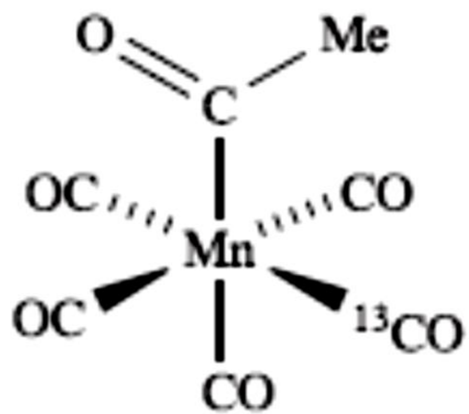
1,1 Insertion

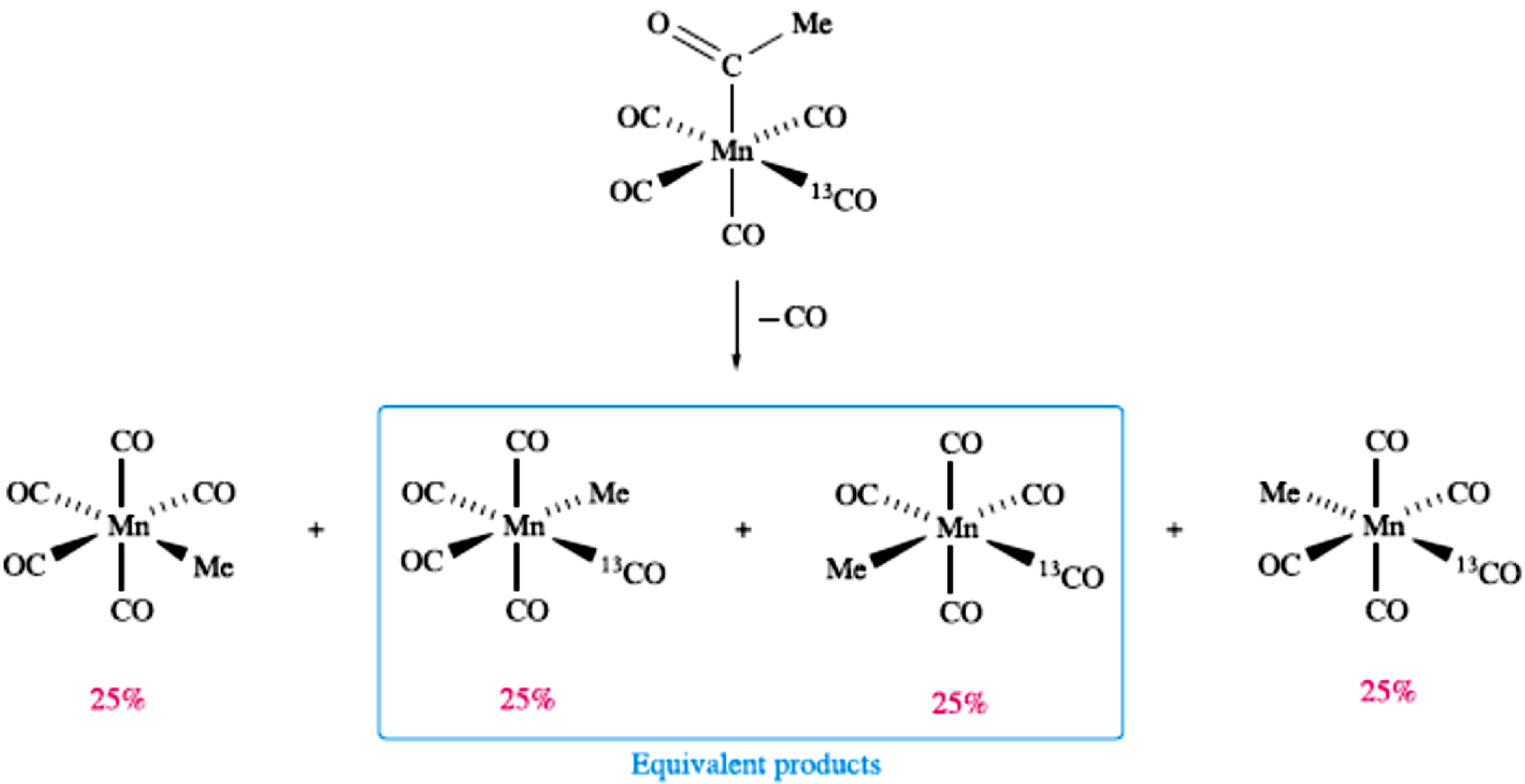


if you only see the result, it *looks like* CO has *inserted* into the M-X bond, hence the name *insertion*

To emphasize that it is actually (mostly) the X group that moves, we use the term *migratory insertion*.


Me
OC
CO
Mn
OC
CO
CO
+ CO
O
Me
C
OC
CO
Mn
OC
CO
CO


O
Me
C
OC
CO
Mn
OC
13CO
CO



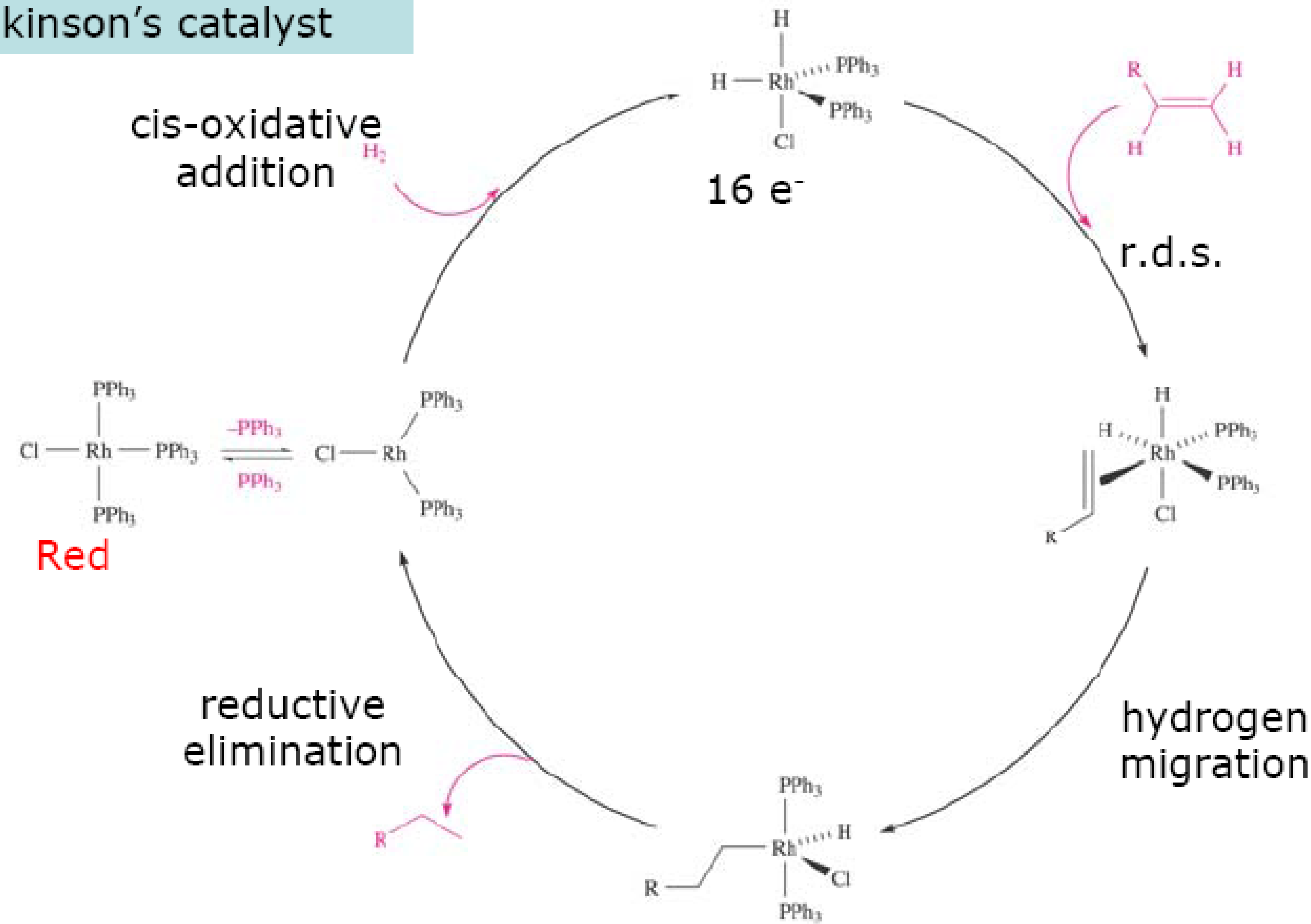
O
Me
C
OC
CO
Mn
OC
^{13}CO
CO
− CO
25%
25%
25%
25%
Equivalent products

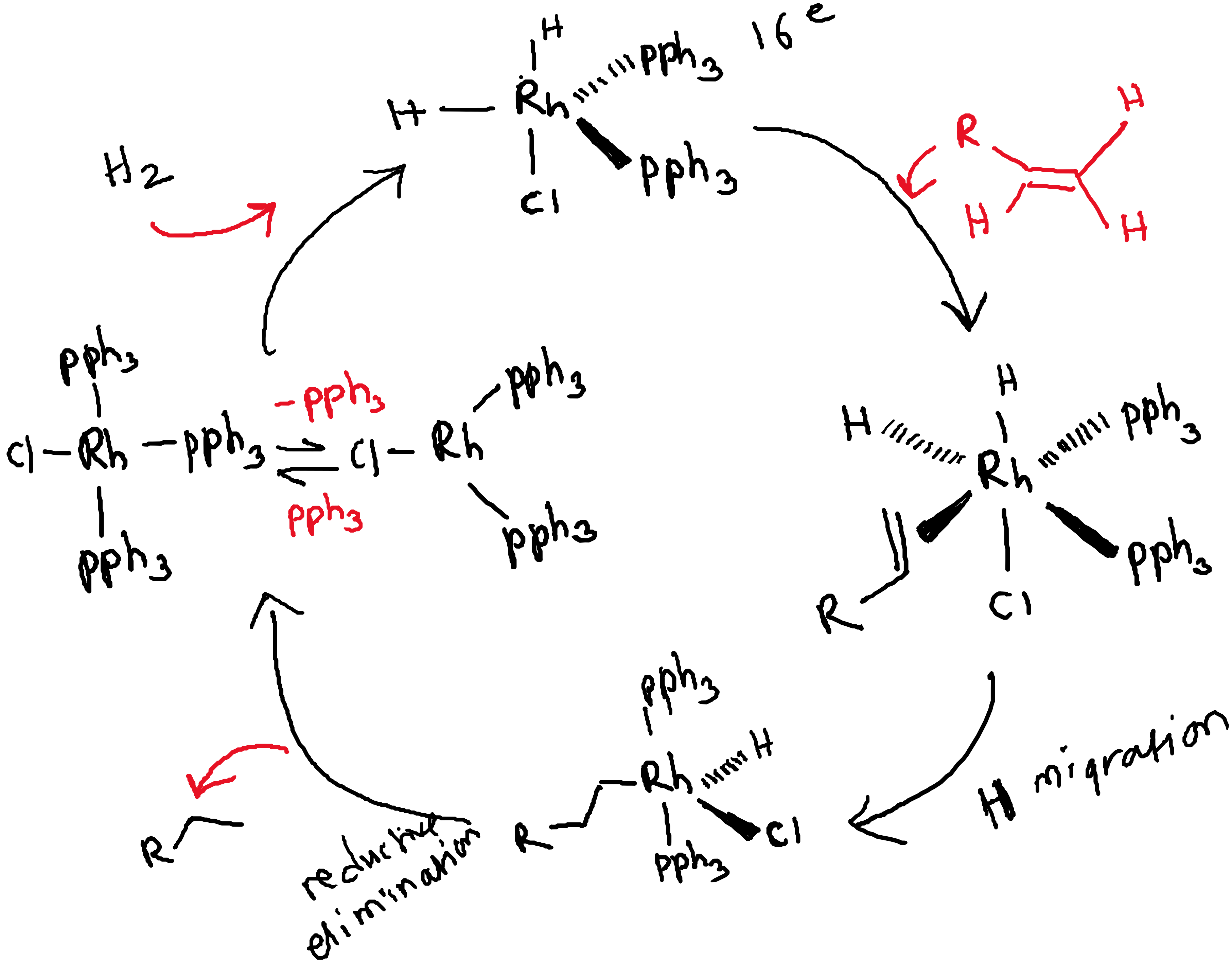
Catalysis (homogeneous)

Alkene Hydrogenation

Catalysis (homogeneous)
Alkene Hydrogenation

Wilkinson's catalyst

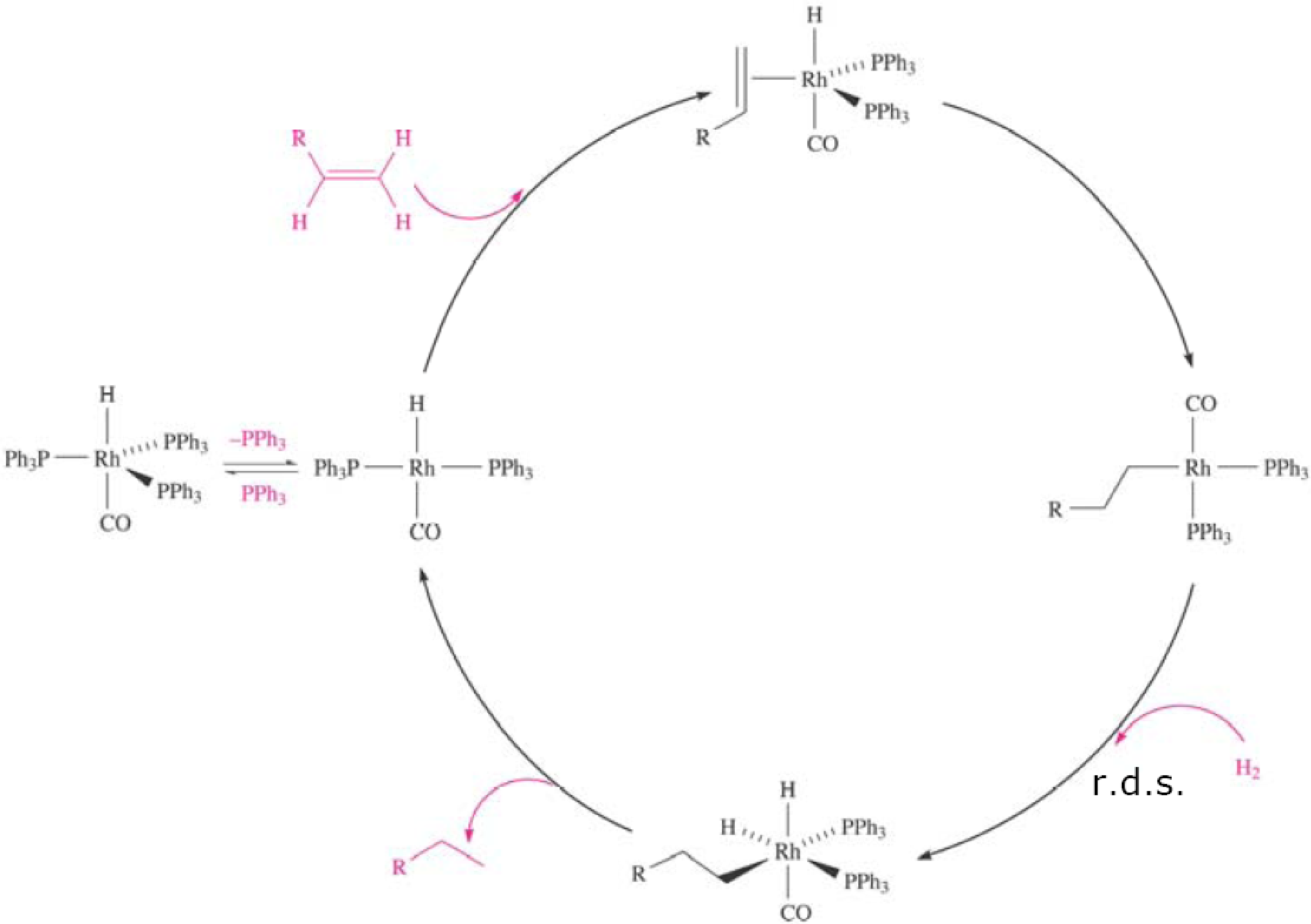




16 e
H2
-PPh3
PPh3
H migration
reductive elimination

The size of the substrate has an effect on the rate of reaction

Same reaction different catalyst



g. 27.8 Catalytic cycle for the hydrogenation of $RCH{=}CH_2$ using $HRh(CO)(PPh_3)_3$ as catalyst.

The Monsanto acetic acid process

Monsanto synthesis

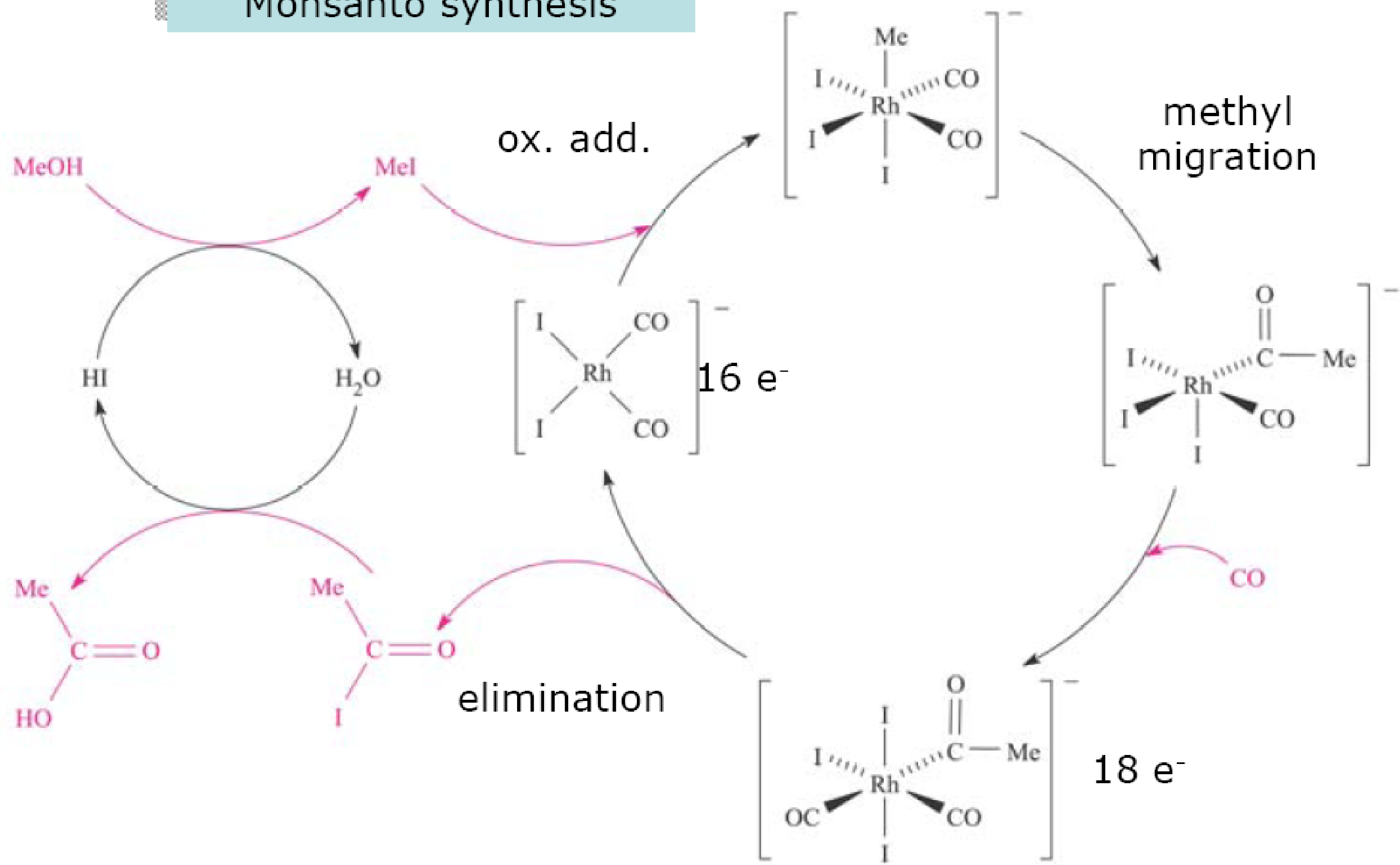
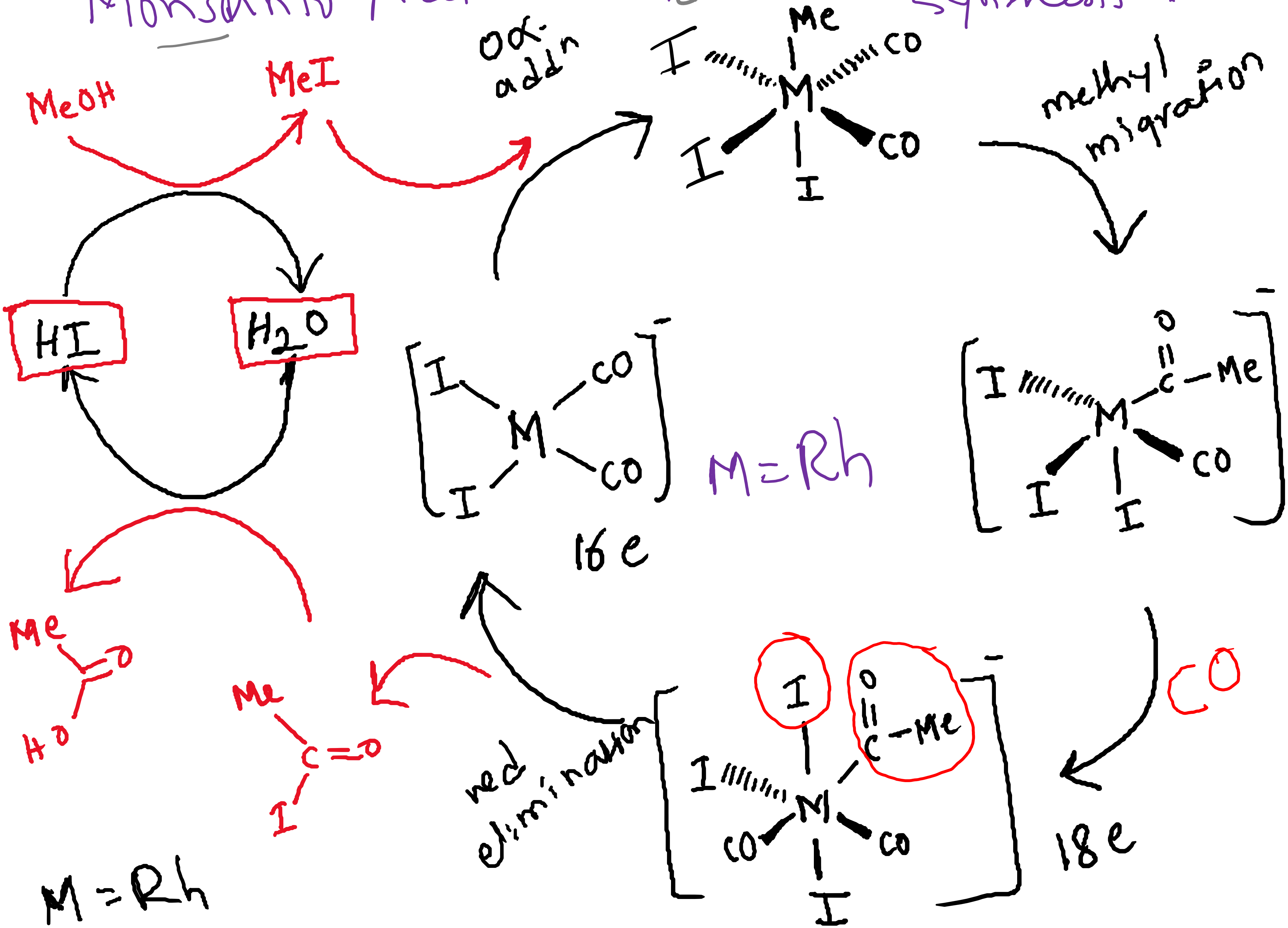


Fig. 27.9 The Monsanto acetic acid process involves two interrelated catalytic cycles.

Monsanto Acetic acid Synthesis :-



The Monsanto acetic acid process

Methanol - reacted with carbon monoxide in the presence of a catalyst to afford acetic acid

Insertion of carbon monoxide into the C-O bond of methanol

The catalyst system - iodide and rhodium

Iodide promotes the conversion of methanol to methyl iodide,

Methyl iodide - the catalytic cycle begins:

1. **Oxidative addition of methyl iodide to $[Rh(CO)_2I_2]^-$**
2. **Coordination and insertion of CO - intermediate 18-electron acyl complex**
3. **Can then undergo reductive elimination to yield acetyl iodide and regenerate our catalyst**

Ziegler-Natta catalysis:

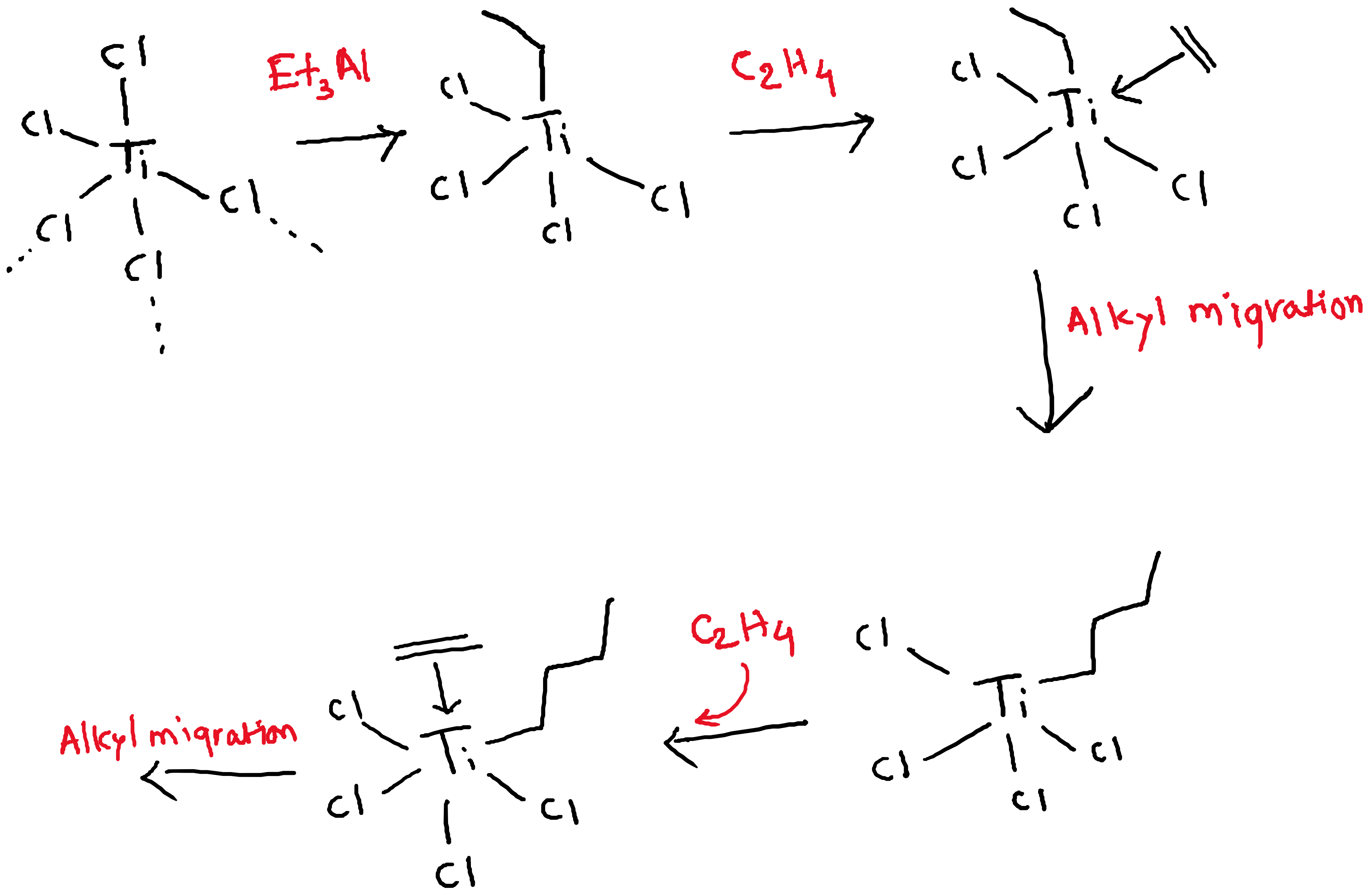
Alkene polymerization

Ziegler-Natta catalysis:

Alkene polymerization

The 1963 the Nobel Prize in Chemistry was awarded to Karl Ziegler and Giulio Natta ‘for their discoveries in the field of the chemistry and technology of high polymers

First generation Ziegler–Natta catalysts were made by reacting $TiCl_4$ with Et_3Al


Et3Al
C2H4
Alkyl migration
C2H4
Alkyl migration
Ti
Cl